Preliminaries

We start by recalling the construction of the group von Neumann algebra. Let G be a countable discrete group. Then, we may form the space

$$\ell^2(G) = \left\{ \left(\xi_g\right)_{g\in G} : \sum_{g\in G} \left|\lambda_g\right|^2 < \infty \right\},$$

with orthonormal basis $\left\{\delta_g\right\}_{g\in G}$. The left regular representation of G is the map $\lambda\colon\ G \to B\left(\ell^2(G)\right)$ that takes $\lambda_g \xi(h) = \xi\left(g^{-1}h\right)$. The *group von Neumann algebra* is then the WOT-closed linear span of $\lambda(G) \subseteq B\left(\ell^2(G)\right)$. We will denote this by $L(G)$. Using the right regular representation (as opposed to the left regular representation) $\rho_g\xi(h) = \xi(hg)$ yields the group von Neumann algebra $R(G)$. We have $R(G) = L(G)'$ and $L(G)' = R(G)$. We will usually work with $R(G)$.

The most important fact about the group von Neumann algebra is that it admits a trace, denoted τ_G, defined by

$$\tau_G(T) = \langle x\delta_e, \delta_e\rangle.$$

We will use this trace as part of the construction of the von Neumann dimension. We note that this trace extends to the space $\mathbb{M}_n(R(G)) = R(G) \otimes \mathbb{M}_n(\mathbb{C})$ via the operator $\tau^{(n)} := \tau \otimes \mathrm{Tr}$, where

$$\tau^{(n)}(x) = \sum_{i=1}^{n} \tau(x_{ii}).$$

We note that the $\ell^2(G)$ can be viewed as a left $\mathbb{C}[G]$-module by using the *right* G-action,

$$\begin{aligned} G \times \ell^2(G) &\to \ell^2(G) \\ (g,\xi) &\mapsto \rho_g\xi. \end{aligned}$$

In particular, this means that we may view $R(G)$ as the space of *left* $\mathbb{C}[G]$-equivariant operators in $B\left(\ell^2(G)\right)$ — recall that ρ_g and λ_g are commutants of each other, so any of the left $\mathbb{C}[G]$ operators will commute with the right action.

Traces of G-module Endomorphisms

Definition: A Hilbert G-module is a Hilbert space V over $\mathbb{C}$ equipped with a $\mathbb{C}$-linear isometric left G-action such that there exists $n \in \mathbb{N}$ and an isometric G-equivariant embedding $V \to \ell^2(G)^{\oplus n}$. Here, it is equivariant with the diagonal left action of G, which we will denote $\lambda^{(n)}$.

Consider a Hilbert G-module H with embedding $i\colon\ H \to \left(\ell^2(G)\right)^n$. We may identify H with a closed G-invariant subspace of $\ell^2(G)^{\oplus n}$. Let $p_{i(H)} \in B\left(\ell^2(G)^{\oplus n}\right)^{\lambda^{(n)}}$ be the orthogonal projection onto $i(H)$. We will let $B(H)^G$ be the set of all G-module endomorphisms of H.

Proposition: We claim that the quantity

$$\dim_{R(G)}(H) = \tau^{(n)}\left(p_{i(H)}\right)$$

is independent of the embedding $i(H)$.

Proof. Suppose $j\colon\ H \to \ell^2(G)^{\oplus m}$ is another embedding with corresponding projection $q_{j(H)}$. Define the map $u\colon \ell^2(G)^{\oplus n} \to \ell^2(G)^{\oplus m}$ by taking $u = j \circ i^{-1}$ on $\mathrm{im}(i)$ and 0 on $\mathrm{im}(i)^{\perp}$. In particular, we have $j = u \circ i$, and $q_{j(H)} = p_{i(H)} \circ u^*$ upon taking adjoints. Therefore, we have

$$\begin{aligned}\tau^{(m)}\left(q_{j(H)}\right) &= \tau^{(m)}\left(j \circ q_{j(H)}\right)\\ &= \tau^{(m)}\left(u \circ i \circ q_{j(H)}\right)\\ &= \tau^{(m)}\left(u \circ i \circ p_{i(H)} \circ u^*\right).\end{aligned}$$

Using the tracial property of $\tau^{(m)}$ and exchanging dimension to account for the new range, we have

$$\begin{aligned}&= \tau^{(n)}\left(i \circ p_{i(H)} \circ u^* \circ u\right)\\ &= \tau^{(n)}\left(i \circ p_{i(H)}^2\right)\\ &= \tau^{(n)}\left(i \circ p_{i(H)}\right).\end{aligned}$$

Therefore, the quantity $\dim_{R(G)}(H)$ is well-defined. □

Remark: For maximum generality, we may consider any linear isometric G-embedding into $\ell^2(G) \otimes K$ for any Hilbert space K, but this will not be an issue we will deal with for a while.

The quantity $\dim_{R(G)}(H)$ is called the von Neumann dimension of H. Certain authors, such as [LÖ2] and [Kam19] define it slightly differently than the way that [LÏ0] does. However, these are equivalent definitions at the end of the day. We note that this quantity is also always positive, since if $P \in \mathbb{M}_n(R(G))$ is the matrix corresponding to the projection $p_{i(H)}$, then all the diagonal entries P_{jj} of P are positive operators, so since $\tau^{(n)}$ is itself positive, $\dim_{R(G)} = \tau^{(n)}\left(p_{i(H)}\right)$ is positive.

Hilbert G-modules are quite like standard $\mathbb{C}[G]$-modules in a variety of ways.

(i) If $K \subseteq H$ is a closed G-invariant subspace of a Hilbert G-module H, then K is itself a G-module, and we call H a submodule.

(ii) If $K \subseteq H$ is a closed G-invariant subspace of a Hilbert G-module H, then H/K is also a Hilbert G-module since we may identify it with the closed G-invariant subspace $K^{\perp} \subseteq H$. We call H/K a quotient module.

(iii) If H_1 and H_2 are Hilbert G-modules with corresponding embeddings $i_1 \colon H_1 \to \ell^2(G)^{\oplus n_1}$ and $i_2 \colon H_2 \to \ell^2(G)^{\oplus n_2}$, then there is an embedding

$$i_1 \oplus i_2 \colon H_1 \oplus H_2 \to \ell^2(G)^{\oplus(n_1+n_2)},$$

yielding the direct sum Hilbert module.

(iv) Let H_1 be a Hilbert G_1-module and H_2 a Hilbert G_2-module with corresponding embeddings $i_1 \colon H_1 \to \ell^2(G_1)^{\oplus n}$ and $i_2 \colon H_2 \to \ell^2(G_2)^{\oplus m}$. These yield a tensor product

$$i_1 \otimes i_2 \colon H_1 \otimes H_2 \to \ell^2(G_1)^{\oplus n} \otimes \ell^2(G_2)^{\oplus m},$$

where we observe that the codomain is isomorphic to $\ell^2(G_1 \times G_2)^{\oplus mn}$. This yields a tensor product Hilbert $(G_1 \times G_2)$-module.

(v) Suppose $G_0 \leq G$ is a finite-index subgroup with index m. There is a functor $\operatorname{Res}_{G_0}^{G}$ from Hilbert G-modules to Hilbert G_0-modules obtained by restricting the group action to G_0.

The corresponding embedding emerges from the fact that taking a system of representatives for the coset space G/G_0 yields a unitary $\operatorname{Res}_{G_0}^{G}\left(\ell^2(G)\right) \cong \ell^2(G_0)^{\oplus m}$. The restriction $\operatorname{Res}_{G_0}^{G}(H)$ is a restricted Hilbert module.

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